

- FRODE ALFSON BJØRDAL, *Visibility Beyond Provability: Converging on Truth*.

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The theory presented is in the tradition of modern type-free truth theories and is developed in a framework that supports a nontrivial set theory beyond Π_1^1 -comprehension; natural strengthenings sufficient to interpret Tarski–Grothendieck set theory are available. A Herzberger–Gupta style semi-inductive definition is presupposed, generating a semantic revision process with a corresponding closure ordinal. The notation $\vdash \ulcorner A \urcorner$ indicates that $\ulcorner A \urcorner$ is a *viewable* theorem; the terminology is apt, for the exclamation mark was long known as the *punctus admirativus* and *signos de admiración* is still a standard term in Spanish. A sentence is viewable if and only if it holds from some ordinal below the closure ordinal onward. A sentence $\ulcorner A \urcorner$ is *visible* just if either $\ulcorner A \urcorner$ or $\ulcorner \neg A \urcorner$ is viewable. Several theorems are *invisible*; salient examples include $\vdash \ulcorner A \urcorner \rightarrow A$, $\vdash \ulcorner A \urcorner \rightarrow \ulcorner \ulcorner A \urcorner \urcorner$, $\vdash \forall x \ulcorner A \urcorner \rightarrow \ulcorner \forall x A \urcorner$, and the new *Planer schema* $\vdash \ulcorner \ulcorner B \urcorner \rightarrow \ulcorner C \urcorner \urcorner \rightarrow \ulcorner B \rightarrow C \urcorner$. Invisible theorems are like eyes: not in the visual field, yet necessary for vision. The distinction between visible theorems and invisible ones corresponds with a truth theoretic distinction in [1, p. 400]. The visibility logic survives Gödel-style diagonalization and contains KD, so it may—given recent literature—be taken to extend Rosserian provability logics. However, it contradicts the logic KDR, proposed as a Rosserian provability logic in [2, p. 598 ff.], whose characteristic axiom is R: $\Box \neg p \rightarrow \Box \neg \Box p$; since extending KDR with the $\Box$ -version $\vdash \Box(\Box B \rightarrow \Box C) \rightarrow \Box(B \rightarrow C)$ of the Planer schema yields $\vdash \Box(\Box A \rightarrow A)$, and inconsistency follows for Montague-style reasons.

[1] ANDREA CANTINI, *Logical Frameworks for Truth and Abstraction*, Studies in Logic and the Foundations of Mathematics, Elsevier, 1996.

[2] TAISHI KURAHASHI, *Rosser Provability and Normal Modal Logics*, *Studia Logica*, vol. 108 (2020), no. 3, pp. 597–617,